



Pitches	50.6%	14.2%	23.3%	57.1%
n = 403	Swing%	Whiff%	CSW%	Zone%

KEY FINDING

Low HAA = 3× more whiffs than High HAA

Low HAA (≈1.08°) generates 25.9% Whiff% — 3× the rate of High HAA (8.5%).

When Beau’s fastball exits the hand at a lower horizontal approach

angle, it arrives from the 1B side and finishes outside to RHH.

That deceptive angle makes it extremely difficult to square up,

driving the dramatic difference in swing-and-miss across folds.

HAA and PlateLocSide share a 0.90 correlation — the approach

angle dictates where the ball ends up, not the other way around.

Low HAA (25.9% Whiff) Mid HAA (11.4% Whiff) High HAA (8.5% Whiff)

TARGET RELEASE POINT

Current RelSide: -2.35 ft

Target RelSide: -2.25 ft

Delta: Δ = +0.10 ft (≈1.2 in toward 1B/glove side)

Expected outcome: Whiff% → ~25.9% (from 14.2% overall)

Moving toward the 1B side of the rubber lowers HAA,

ball arrives from 1B side → stays outside vs RHH

— the most effective approach angle.

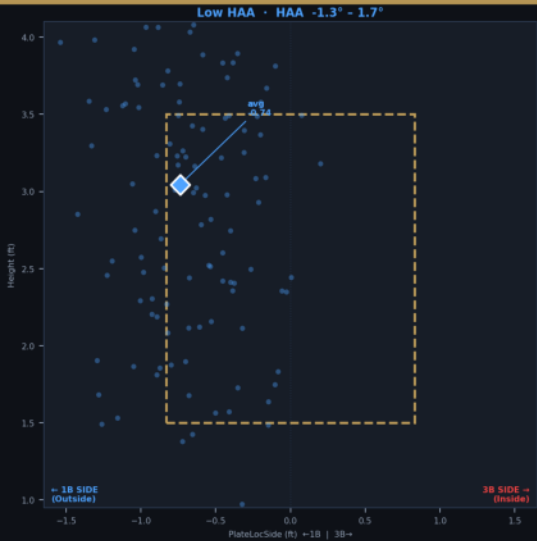
Guardrail: Zone% drops to 41.5% in Low-HAA fold.

Pair with location discipline or pitch-to-contact mix.

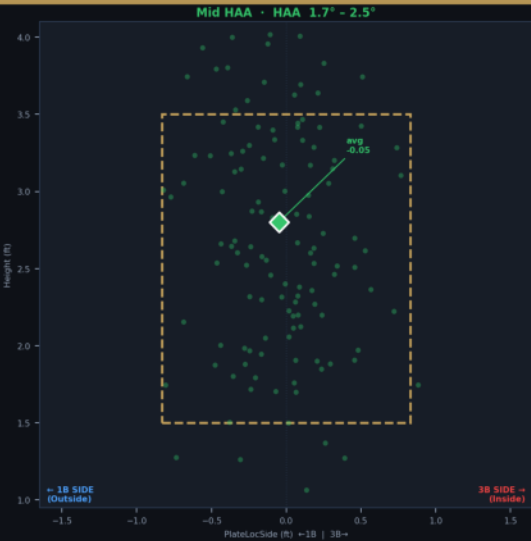


Elson, Beau (LHP) — Fastball HAA Fold Analysis · vs RHH Only

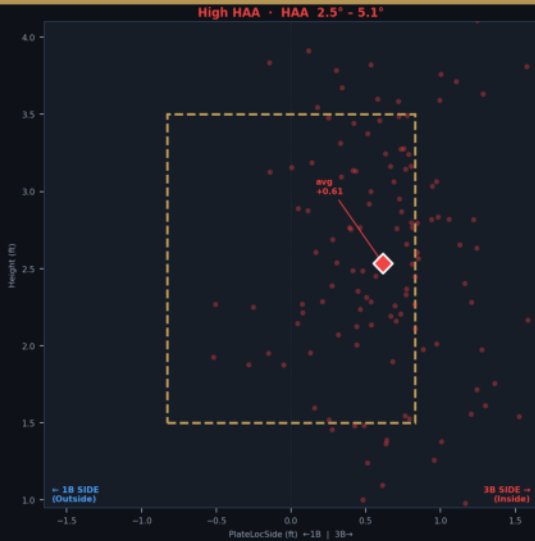
Fastballs split into 3 equal HAA terciles · Positive HAA = ball arriving from 3B side (inside to RHH) · Negative/low HAA = ball arrives from 1B side (outside to RHH) · Low HAA most effective



Outside (1B side)	
N (vs RHH)	135
Avg HAA	1.08°
Avg Velo	89.3 mph
Avg Rel Side	-2.25 ft
Avg Location	-0.74 ft
Strike%	57.8%
Zone%	41.5%
CSW%	28.1%
Whiff%	25.9%
Ball stays 1B-side / away from RHH → Most whiffs, highest CSW	

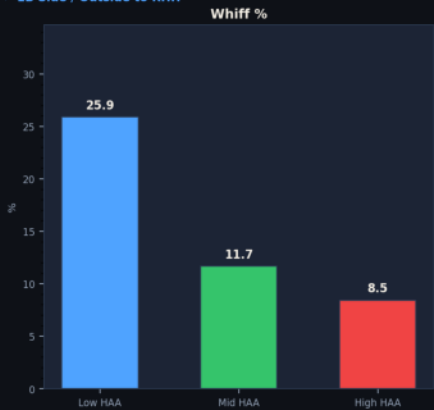


Middle	
N (vs RHH)	134
Avg HAA	2.12°
Avg Velo	89.2 mph
Avg Rel Side	-2.35 ft
Avg Location	-0.05 ft
Strike%	75.4%
Zone%	73.9%
CSW%	23.1%
Whiff%	11.7%
Ball works middle of plate → Moderate outcomes	

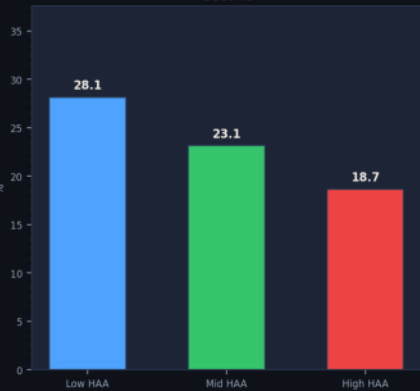


Inside (3B side)	
N (vs RHH)	134
Avg HAA	3.09°
Avg Velo	88.9 mph
Avg Rel Side	-2.44 ft
Avg Location	+0.61 ft
Strike%	67.2%
Zone%	55.2%
CSW%	18.7%
Whiff%	8.5%
Ball runs inside (3B-side) to RHH → Harder to generate whiffs	

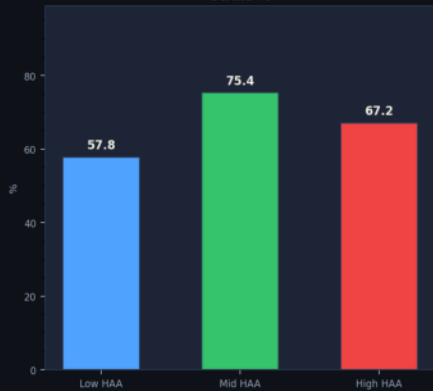
→ 1B Side / Outside to RHH



CSW %

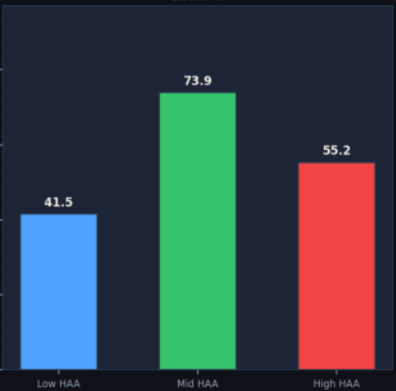


Strike %



→ 3B Side / Inside to RHH

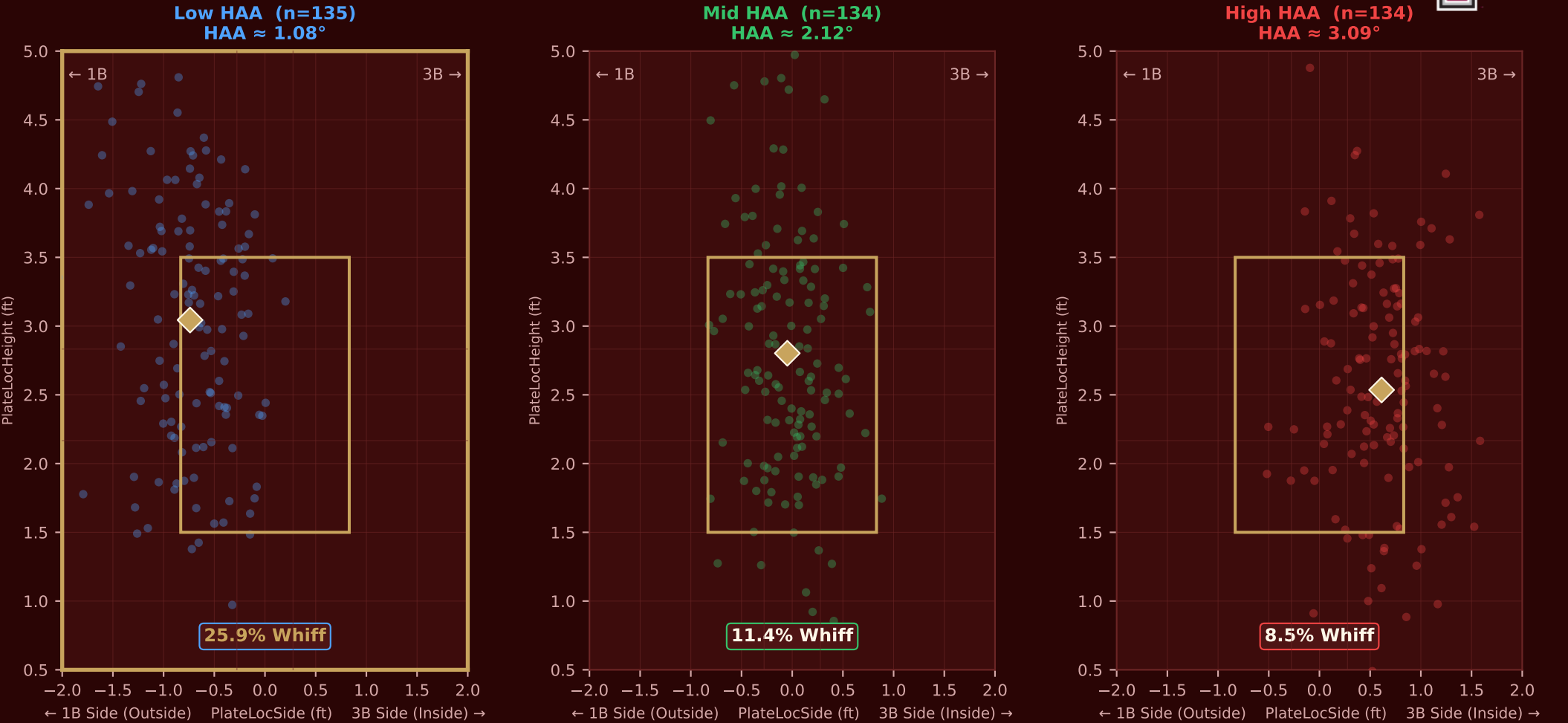
Zone %



Low HAA · Outside (1B side) · HAA -1.3° - 1.7° Mid HAA · Middle · HAA 1.7° - 2.5° High HAA · Inside (3B side) · HAA 2.5° - 5.1°

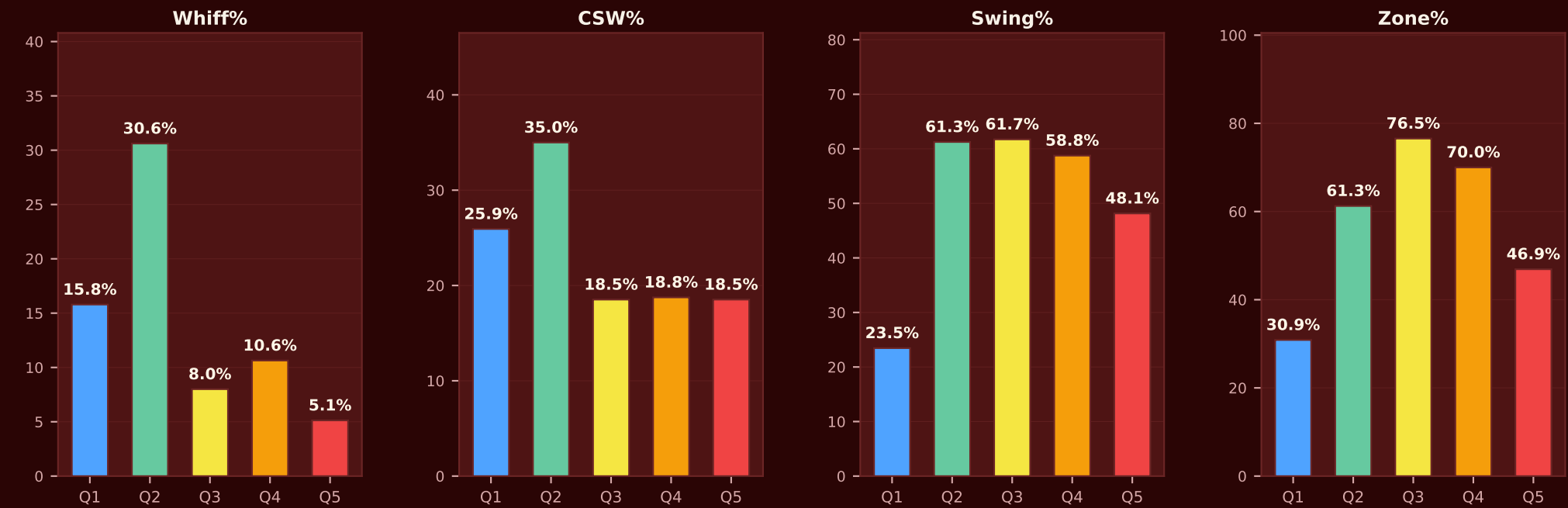
Three HAA tercile folds (Low / Mid / High) vs RHH | Low fold: HAA ≈ 1.08° → 25.9% Whiff% · Mid fold: HAA ≈ 2.12° → 11.4% Whiff% · High fold: HAA ≈ 3.09° → 8.5% Whiff%

Beau Elson · Zone Location by HAA Fold (vs RHH)



Fold Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Low	135	1.08°	-2.25	5.66	40.0%	25.9%	28.1%	41.5%	35.7%	64.3%	22.8%	83.3%	55.6%
Mid	134	2.12°	-2.35	5.61	59.0%	11.4%	23.1%	73.9%	26.3%	73.7%	17.1%	90.4%	66.7%
High	134	3.09°	-2.44	5.58	53.0%	8.5%	18.7%	56.0%	24.0%	76.0%	23.7%	94.7%	78.6%

Beau Elson · 5-Fold HAA Analysis (Quintiles vs RHH)



5-Fold (Quintile) Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Q1 Low	81	0.78°	-2.25	5.68	23.5%	15.8%	25.9%	30.9%	52.0%	48.0%	12.5%	91.7%	71.4%
Q2	80	1.62°	-2.28	5.63	61.3%	30.6%	35.0%	61.3%	30.6%	69.4%	48.4%	79.4%	46.7%
Q3	81	2.12°	-2.35	5.61	61.7%	8.0%	18.5%	76.5%	21.0%	79.0%	5.3%	91.8%	100.0%
Q4	80	2.58°	-2.38	5.60	58.8%	10.6%	18.8%	70.0%	23.2%	76.8%	16.7%	90.7%	75.0%
Q5 High	81	3.37°	-2.47	5.57	48.1%	5.1%	18.5%	46.9%	26.3%	73.7%	25.6%	100.0%	81.8%

Q1 (lowest HAA, avg 0.78°) → outside to RHH Q5 (highest HAA, avg 3.37°) → inside to RHH